

AP CSP | Lesson 14: Conditional Loops

Homework | Multiple-choice and written practice

Name: _____ Class: _____ Date: _____

Time: 45 minutes | Total: 30 teacher marks (not an AP score) | Show working.

Original AP-style MCQs and classroom written practice; these are not official AP questions.

12 original questions, 30 teacher marks, about 45 minutes. Q5 requires exactly TWO choices. Show predictions, traces and reasons. Braces mark blocks; each statement is on its own line.

1. A REPEAT UNTIL condition is true before the first check. How many times does its body run? [1 mark]

A. 0 B. 1 C. 2 D. Infinitely many

Answer: _____

2. Start $n = 2$. While the stop condition $n \geq 7$ is false, add 2 to n . What is the final n ? [1 mark]

A. 6 B. 7 C. 8 D. 10

Answer: _____

3. Start $x = 1$. A REPEAT UNTIL($x = 8$) loop only sets $x \leftarrow x + 2$. What happens? [1 mark]

A. It stops at 7. B. It stops at 8.
C. It stops at 9. D. Its condition never becomes true.

Answer: _____

4. A loop sums nonnegative inputs until -1 . Why should it check $\text{value} = -1$ before adding value? [1 mark]

A. To add -1 exactly once. B. To exclude the sentinel from the total.
C. To discard all zero inputs. D. To read every value twice.

Answer: _____

5. Which TWO changes can prevent this loop from ending? Start $\text{count} = 0$; stop when $\text{count} \geq 3$; body displays count and adds 1. [1 mark]

A. Remove the update. B. Start count at 3.
C. Subtract 1 instead of adding 1. D. Change the stop test to $\text{count} \geq 2$.

Answer: _____

6. Which loop most directly expresses “perform exactly five trials”? [1 mark]

A. REPEAT UNTIL(trials = 0), with no update B. REPEAT UNTIL(true)
C. REPEAT 5 TIMES D. REPEAT UNTIL(trials > 5), starting at 0 and adding 1

Answer: _____

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Written responses | Questions 7-9

7.

[4 marks]

```
n ← 1
REPEAT UNTIL(n ≥ 6)
{
  DISPLAY(n)
  n ← n + 2
}
```

Give (a) displayed values, (b) final n, (c) body count, (d) condition-check count.

8.

[4 marks]

Start charge at 20 and add 15 each run. Stop when charge is at least 65.

(a) Write initialization. (b) Write REPEAT UNTIL with its body.

(c) Give the final charge. (d) Give the number of body runs.

9.

[4 marks]

Start x = 2. Stop at x ≥ 5. The body displays x but never changes it.

(a) Explain the failure. (b) Give a repair after DISPLAY.

(c) Give the repaired output. (d) Give an initially true test and its body count.

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Written responses | Questions 10-12

10. [4 marks]

A correct sentinel loop initializes $\text{total} = 0$, reads value, and until $\text{value} = -1$ adds value then reads again. Inputs are 5, 0, -1.

(a) Give final total. (b) Give body count. (c) Give INPUT-call count. (d) Explain why 0 and -1 are treated differently.

11. [4 marks]

A model starts $n = 1$ and adds 2 repeatedly. Compare stop tests $n = 6$ and $n \geq 6$.

(a) Does equality terminate? Explain. (b) Does $\geq$ terminate? Give final n .

(c) Give the $\geq$ loop's body count. (d) State a requirement that makes $\geq$ suitable.

12. [4 marks]

Express "display Ready exactly three times" using REPEAT UNTIL and count.

Give (a) initialization, (b) stopping condition, (c) complete body, and (d) a count trace proving three executions.
